

$$1. \begin{cases} \frac{\partial u}{\partial t} = \Delta u \\ u|_{t=0} = \varphi(x) \end{cases} \quad \text{注意完整版} \quad \begin{cases} \frac{\partial u}{\partial t} = d^2 \nabla^2 u - \beta(u - u_{env}) \\ -k \frac{\partial u}{\partial n} \Big|_{\partial \Omega} = h(u - u_{env}) \text{ 边界} \end{cases}$$

$$(2) u_{xy} = xy \Leftrightarrow u = u_2 + u_1 \quad u_1 = \frac{x^2 y^2}{2!} \quad u_{1xy} = xy$$

$$\Rightarrow u_{2xy} = 0 \Rightarrow u_2(x, y) = g(x) + f(y)$$

$$\Rightarrow u_{xy} = xy + g(x) + f(y)$$

$$(3) 1. f(t, x) = 0 \quad \text{达朗贝尔... 公式}$$

$$\Rightarrow \frac{1}{2} (x - \sqrt{2}t + x + \sqrt{2}t) + \frac{1}{2\sqrt{2}} \int_{x-\sqrt{2}t}^{x+\sqrt{2}t} 2 \sin s \, ds$$

$$= x + \frac{1}{\sqrt{2}} (\cos(x - \sqrt{2}t) - \cos(x + \sqrt{2}t))$$

$$2. f(t, x) = 3e^t \sin x$$

$$f_1 = ae^t \sin x \quad u = f_1 + u_1$$

$$f_1(t) = 2f_1(x) = 3e^t \sin x$$

$$\Rightarrow a = -1$$

$$\Rightarrow u = -e^t \sin x + u_1$$

$$\begin{cases} u_1(t) = 2 & u_{1xx} \end{cases}$$

$$\begin{cases} u_1(t=0) = x + \sin x & u_1|_{t=0} = 3 \sin x \end{cases} \quad \text{由达朗贝尔}$$

$$\Rightarrow u_1 = x + \frac{1}{2} \sin(x - \sqrt{2}t) + \frac{1}{2} \sin(x + \sqrt{2}t)$$

$$+ \frac{3}{2\sqrt{2}} (\cos(x - \sqrt{2}t) - \cos(x + \sqrt{2}t))$$

$$2. \begin{cases} \lambda > 0 \\ \lambda < 0 \\ \lambda = 0 \end{cases}$$

$$\textcircled{2} \lambda < 0 \quad \lambda = -v^2$$

$$\Rightarrow Ae^{-vx} + Be^{vx}$$

$$Y'(0) = 0 \Rightarrow -Av + Bv = 0$$

$$A=B$$

$$Y(5) = 0 \quad Ae^{-5v} + Be^{5v} = 0$$

$$\Rightarrow A=B=0$$

$$\textcircled{3} \lambda = 0 \Rightarrow Ax + B$$

$$x \neq 0 \quad Y'(0) = 0 \Rightarrow A = 0$$

$$Y(5) = 0 \Rightarrow B = 0 \quad (\text{舍})$$

$$\textcircled{1} \lambda > 0 \quad \lambda = v^2$$

$$Y = A \sin vx + B \cos vx \quad Y'(0) = 0$$

$$\Rightarrow A = 0 \quad Y(5) = 0$$

$$5v = (n + \frac{1}{2})\pi \quad n \in \mathbb{Z}$$

$$\Rightarrow \lambda = \frac{(n + \frac{1}{2})^2}{25} \pi^2 \quad \text{解}$$

$$A_n \cos(n + \frac{1}{2})\pi x$$

$$(2) \quad \text{由题意上一问} \quad \text{分离变量}$$

$$\frac{T''}{T} = \frac{X''}{X} + \pi^2 = -\lambda$$

$$U_X(0) = U_X(1) = 0$$

$$X'' + \lambda_1 X = 0$$

同上且 $\lambda > 0$ $v^2 = \lambda$
 $X = A \sin vx + B \cos vx$
 $X' = -Av \sin vx + Bv \cos vx$

$\Rightarrow v = n\pi$

$\Rightarrow \lambda_1 = (n\pi)^2$
 $\lambda + \pi^2 = (n\pi)^2$

$\lambda = (n\pi)^2 - \pi^2$

故而 $\frac{T''}{T} = -(n^2-1)\pi^2$ $\pi^2(n^2-1) = s_n^2$

故而解为 $A_0 e^{-\pi t} + B_0 e^{\pi t} + (A_1 t + B_1) \cos \pi x$
 $+ \sum_{n=2}^{+\infty} (A_n \cos s_n t + B_n \sin s_n t) \cos n\pi x$
 总结原式由正交性

$\Rightarrow \begin{cases} A_0 + B_0 = 0 \\ -\pi A_0 + B_0 \pi = 1 \end{cases} \quad \begin{cases} B_1 = 1 \\ A_1 = 0 \end{cases}$
 $\Rightarrow B_0 = \frac{1}{2\pi} \quad A_0 = -\frac{1}{2\pi}$
 $\begin{cases} A_2 = 0 \\ \sin B_2 = 1 \end{cases} \quad \Rightarrow B_2 = \frac{1}{\sqrt{3}\pi}$

$\Rightarrow$ 原式 $= \frac{1}{2\pi} (e^{\pi t} - e^{-\pi t}) + \cos \pi x + \cos 2\pi x (\sin \sqrt{3}\pi t) \frac{1}{\sqrt{3}\pi}$

分离变量

三. 注意与 0 无关 轴对称

$\frac{T'}{T} = \frac{R''}{R} + \frac{1}{r} \frac{R'}{R} = -\lambda$

$\lambda = \omega^2$

则 R 解为

$J_0(\omega r)$

其中 $J_0(2\omega) = 0$

故而

$\sum_{n=1}^{+\infty} \frac{J_0(\omega_n r)}{\omega_n} e^{-\omega_n^2 t} A_n$
 ω_n 为

$J_0(2\omega_n) = 0$ 由 β_n
 求解

三、求 又. $u|_{t=1} = 4$

$$\Rightarrow A_n e^{-w_n^2} = \frac{\int_0^2 r \cdot J_0(w_n r) dr \cdot 4}{|| J_0^2(w_n r) ||}$$

$$= \frac{\int_0^{2w_n} t J_0(t) dt \cdot 4}{w_n^2 \frac{4}{2} J_1^2(2w_n)}$$

$$= \frac{2w J_1(2w)}{w^2 \frac{4}{2} J_1^2(2w)}$$

$$= \frac{1}{w J_1(2w)}$$

$$\Rightarrow A_n = e^{w_n^2} \frac{1}{w J_1(2w)}$$

故可得

$$\sum_{n=1}^{+\infty} \frac{e^{w_n^2}}{w J_1(2w)} J_1(wr) e^{-w_n^2 t}$$

其中 w_n 为 $J_0(2w_n) = 0$ 第 n 个根

四. 注意与 Q 无关, 轴对称.

设 $u = u(r, \theta)$

注意 等轴对称
入特征正为 $P_n(\cos \theta)$

$$\sum_{n=0}^{+\infty} (A_n r^{-n-1} + B_n r^n) P_n(\cos \theta)$$

$$\begin{cases} \pm \infty \\ -0 \end{cases} \lim_{r \rightarrow \infty} \text{有限} \quad \lim_{r \rightarrow +\infty} u = 1$$

$$\Rightarrow \text{原式} = A_n r^{-1} + 1 + \sum_{n=1}^{+\infty} r^{-n-1} P_n(\cos \theta) A_n$$

$$\text{又 } u|_{r=2} = \cos^2 \theta + \cos \theta + 1$$

$$= \frac{2}{3} P_2(\cos \theta) + \frac{4}{3} P_0(\cos \theta) + P_1(\cos \theta)$$

由 $\frac{2}{3} + \frac{4}{3} + 1$

$$\Rightarrow \text{原式} = 1 + 2r^{-2} P_1(\cos \theta) + r^{-3} \frac{16}{3} P_2(\cos \theta)$$

II. 对原式 $\times$ Fourier 变换 $F(u) = \bar{U}$

$$\frac{d\bar{U}}{dt} = -4\lambda^2 \bar{U} - i\lambda \bar{U} + \bar{U}$$

$$\bar{U}_{t=0} = \overline{\varphi(x)}$$

故有 $\bar{U} = \overline{\varphi(x)} e^{(-4\lambda^2 - i\lambda + 1)t}$

(1) $F(\delta(x)) = 1 \Rightarrow u = F^{-1}[\bar{U}]$
 $= F^{-1}[e^{(-4\lambda^2 - i\lambda + 1)t}]$

$$= e^t f(x+t)$$

$$f(x) = \frac{1}{4\sqrt{\pi t}} \exp\left(-\frac{x^2}{16t}\right)$$

$$\Rightarrow u = e^t \frac{1}{4\sqrt{\pi t}} \exp\left(-\frac{(x+t)^2}{16t}\right)$$

$$(2) \int_{-\infty}^{+\infty} \sin x e^{-i\lambda x}$$

$$= \int_{-\infty}^{+\infty} \frac{e^{ix} - e^{-ix}}{2i} e^{-i\lambda x} dx$$

$$= \int_{-\infty}^{+\infty} \frac{e^{-ix(\lambda-1)} - e^{-ix(\lambda+1)}}{2i} dx$$

$$= \frac{\delta(\lambda-1) - \delta(\lambda+1)}{2i}$$

$$= \pi \frac{\delta(\lambda-1) - \delta(\lambda+1)}{i}$$

故有 F^{-1}

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{\pi (\delta(\lambda-1) - \delta(\lambda+1))}{i} e^{(-4\lambda^2 - i\lambda + 1)t} e^{i\lambda x} dx$$

$$\begin{aligned}
 & \frac{1}{2i} \left(e^{(-4-i+1)t} e^{ix} - e^{(-4+i+1)t} e^{-ix} \right) \\
 &= e^{-3t} \frac{1}{2i} \left(e^{(x-t)i} - e^{(t-x)i} \right) \\
 &= e^{-3t} \sin(x-t)
 \end{aligned}$$

六. (1) Green 函数 $\Delta G = -\delta$ 通解 为 $\Delta G = -\delta$

$$\begin{cases} \Delta G = -\delta \\ G|_{\partial D} = 0 \end{cases} \quad \frac{1}{2\pi} \ln \frac{1}{r}$$

$\Rightarrow M_0(x_0, y_0) \quad M_1(-8-x_0, y_0)$

解: $G = \frac{1}{2\pi} \ln \frac{(x+8+x_0)^2 + (y-y_0)^2}{(x-x_0)^2 + (y-y_0)^2}$
(M, M_0).

$$x_1 = \frac{x}{2} \quad y_1 = \frac{y}{3}$$

原式 $\Rightarrow$

$$x_1 > -2$$

$$\begin{cases} u_{x_1 x_1} + u_{y_1 y_1} = \frac{1}{6} \delta(x_1, y_1) \\ u|_{x_1 = -2} = \varphi(y_1) \end{cases}$$



故而原式 =

$$\iint_{\mathbb{R}^2} -\frac{1}{6} \delta(x_1, y_1) G(M, M_0) dM_0$$

$$- \int \varphi(y_1) \left(\frac{\partial G}{\partial x_0}, \frac{\partial G}{\partial y_0} \right) (-1, 0) dy_0$$

$$\frac{\partial G}{\partial x_0} = \frac{1}{4\pi} \frac{(x_0 + 4 + x')^2 (x - x_0)^2 + (y - y_0)^2 - (x_0 - x')^2 + (y' - y)^2}{((x' - x_0)^2 + (y' - y_0)^2)^2} \quad x_0 = -2$$

$$= \frac{1}{4\pi} \frac{2(2+x)}{(x'+2)^2 + (y'-y_0)^2}$$

$$= \frac{2+x'}{\pi(x'+2)^2 + (y'-y_0)^2}$$

原式 $= -\frac{1}{6} \frac{1}{4\pi} \ln \frac{(x'+4)^2 + y'^2}{x'^2 + y'^2} \quad \begin{matrix} y' = \frac{y}{3} \\ x' = \frac{x}{2} \end{matrix}$

$$+ \int \varphi(3y_1) \frac{2+x'}{\pi[(x'+2)^2 + (y'-y_1)^2]} dy_1$$

$$= \frac{1}{\pi} \left[\int_{-\infty}^{+\infty} \varphi(3y_1) \frac{2 + \frac{x}{2}}{\left(\frac{x}{2} + 2\right)^2 + \left(\frac{y}{3} - y_1\right)^2} dy_1 \right]$$

$$-\frac{1}{24} \ln \left(\frac{\left(\frac{x}{2} + 4\right)^2 + \frac{y^2}{9}}{\frac{x^2}{4} + \frac{y^2}{9}} \right) \quad]$$